

KEVIN OBOTE

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PROFESSIONAL SUMMARY

Dedicated Research Scholar with a BSc (Upper Second-Class Honors) in Mathematics and Computer Science and an ongoing MSc in Data Science and Analytics at Strathmore University. Proven track record in conducting high-impact lab experimentation, sentiment analysis, and NLP research. Recently ranked **#2 in Africa** in the Deep Learning Indaba Community Challenge and recognized as a **Data Science Africa (DSA) 2025 Hackathon Finalist**. Expert in data analytics, proposal development for societal challenges, and student mentorship within technical environments.

EDUCATION

Strathmore University - Nairobi, Kenya
MSc. Data Science and Analytics

May 2024 - Present

ALX – Nairobi, Kenya
Data Science

May 2023 – July 2024

Machakos University – Machakos, Kenya
BSc. Mathematics and Computer Science

September 2017 – April 2023

TECHNICAL SKILLS

- **Languages:** C, Python, SQL, Java, R, JavaScript, HTML/CSS
- **Data Engineering:** Hadoop, Spark, Azure, NoSQL, PostgreSQL
- **Research Expertise:** Generative AI, Natural Language Processing (NLP), Deep Learning Architectures (Transformers, CNNs, RNNs), Ethical AI, Computer Vision.
- **Advanced Tools & Libraries:** PyTorch, TensorFlow, Hugging Face (for Transformers), spaCy, OpenCV.
- **DevOps & APIs:** Git, Postman, REST APIs, Laravel, FAST API
- **Domains:** Agritech, Healthcare, NLP, Elections, Speech Tech
- **Networking:** Devnet, TCP/IP, DHCP
- GTFS data analysis and optimization.
- VertexAI, Agent Development Kit

MATHEMATICAL FOUNDATION

- **Pure Mathematics:** Complex Analysis, Real Analysis, Abstract Algebra, Functional Analysis, Topology, Differential Equations
- **Probability & Statistics:** Linear Algebra, Calculus, Probability Theory, Bayesian Statistics, Causal Inference

RESEARCH EXPERIENCE

Research Scholar

May 2024 - Present

iLabAfrica – Nairobi, Kenya

- Spearheaded predictive model development using PySpark for large-scale analytics, focusing on novel model architectures for time-series data
- Engineered and automated ML pipelines for real-time model scoring and performance monitoring.
- Contributed to foundational research on ethical AI principles and their application in policy development.
- Authored and published two technical reports on AI applications in natural resource management.

Research Enumerator

Jan 2025 - Feb 2025

Nutritional International – Nairobi, Kenya

- Facilitated nutritional endline survey in Kiambu County.
- Data survey and questionnaire review.
- Trained community health workers and improved survey response.

- Architected and developed a Swahili speech-to-text NLP model, achieving over 85% accuracy. The model was open-sourced by being pushed to Hugging Face, significantly enhancing regional accessibility and collaborative development.
- Currently leading the research and development of a generative, multilingual chatbot designed for mental health and agribusiness applications, with a focus on code-switching between English, Kiswahili, and Sheng.
- Pioneering the development of Sema Sasa, an automatic speech recognition (ASR) solution aimed at promoting inclusivity in education by serving as a learning assistant. This project is currently in progress.
- Contributed to the full development lifecycle, from foundational research to documenting workflows and communicating technical findings to non-technical stakeholders.
- Designed and deployed analytics dashboards for internal and client-side usage, leveraging data to inform strategic decision-making.

PROFESSIONAL EXPERIENCE

Associate Product Manager
eProd Solutions Limited – Nairobi, Kenya

May 2023 - June 2024

- Developed and integrated a novel ML-driven forecasting model into an ERP system, improving supply chain predictions by 20% by leveraging deep learning architectures.
- Designed and deployed Power BI dashboards for over 20 agribusiness clients, enhancing client reporting speed by 40%.
- Managed 3 product interns and initiated a data literacy program, reducing onboarding time by 30%.

Election Project Manager (Data & System Ops)
Roman Solutions Limited – Machakos, Kenya

February 2022 - March 2023

- Designed a secure, online university voting system used by over 7,000 students..
- Reduced voting time by 60% and improved participation by 35% through system UX enhancements
- Implemented real-time monitoring for ballot integrity and fraud detection.

EARLY CAREER EXPERIENCE

- **Data Analyst & Client Officer** – eProd Solutions (Sept 2022 – May 2023)
- **Software Engineer Intern** – eProd Solutions (Jan 2022 – Feb 2023)
- **IT Intern** – Swahilipot Hub Foundation (Jan 2021 – Apr 2021)

PROJECT HIGHLIGHTS

- **Swahili Speech-to-Text Model:** Pioneered the development of a Swahili-specific speech-to-text model. This involved data synthesis, pre-processing, and fine-tuning a transformer-based architecture to achieve a Word Error Rate (WER) of 15%, significantly improving accessibility and NLP capabilities for the region. This project demonstrates your ability to conduct independent research, work with limited resources, and contribute to a specific, impactful area of AI.
- **Kenya Natural Resource Policy Research** – Data-driven policy recommendations with spatial analysis.
- **Nutritional Survey Automation** – Data collection and analysis pipeline for public health insight.

REFERENCES

Dr. Evans Omondi Associate Research Scientist APHRC, Nairobi Kenya eomondi@strathmore.edu +254735743609	Dr. Betsy Muriithi Research Fellow iLabAfrica, Strathmore University bmuriithi@strathmore.edu +254750941086	Dr.Benjamin Kikwai Lecturer : Pure Mathematics Machakos University kikwaib@mksu.ac.ke +254715362549
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